Atlas V1 Static Torque Analysis

1. Objective

The objective of this analysis is to determine whether the selected MG995 shoulder servo can statically support the Atlas V1 robotic arm under its maximum collision-limited extension. The required shoulder holding torque is calculated using SolidWorks mass properties and compared against the manufacturer's specified stall torque.

The analysis estimates the torque required at both the shoulder and elbow joints when the arm is positioned at its maximum practical extension (collision-limited configuration). The calculated torque will then be compared to the manufacturer's rated torque of the MG995 servo.

2. Assumptions

The following assumptions are made:

- Static analysis only (no acceleration).
- Gravity acts vertically downward.
- All printed components are modeled using Anycubic PLA V3.0 material.
- CAD mass properties obtained from SolidWorks are assumed accurate.
- Servo masses are taken from manufacturer specifications.
- Wiring, screws, bearings, and payload are neglected during the initial analysis.
- The arm configuration corresponds to the maximum extension achievable before self-collision.

3. Component Masses

Component	**Mass (g)**	**Mass (kg)**
Link 1 Printed Structure	280.24	0.28024
Elbow MG995 Servo	55	0.05500
Link 2 Printed Structure	164.24	0.16424
Wrist MG90S Servo	13.4	0.01340
Printed End Effector	30.23	0.03023
Gripper MG90S Servo	13.4	0.01340

4. Weight Calculation

Each component weight is calculated using

$$W = mg$$

where

- m= component mass (kg)
- $g = 9.81\ m/s^2$

Weight Calculations

Link 1

$$W_1 = 0.28024(9.81)$$
$$W_1 = 2.749N$$

Elbow Servo

$$W_2 = 0.055(9.81)$$
$$W_2 = 0.540N$$

Link 2

$$W_3 = 0.16424(9.81)$$
$$W_3 = 1.611N$$

Wrist Servo

$$W_4 = 0.0134(9.81)$$
$$W_4 = 0.131N$$

Printed End Effector

$$W_5 = 0.03023(9.81)$$
$$W_5 = 0.297N$$

Gripper Servo

$$W_6 = 0.0134(9.81)$$
$$W_6 = 0.131N$$

Summary

Component	Mass (g)	Mass (kg)
Link 1 Printed Structure	280.24	0.28024
Elbow MG995 Servo	55	0.05500
Link 2 Printed Structure	164.24	0.16424
Wrist MG90S Servo	13.4	0.01340
Printed End Effector	30.23	0.03023
Gripper MG90S Servo	13.4	0.01340

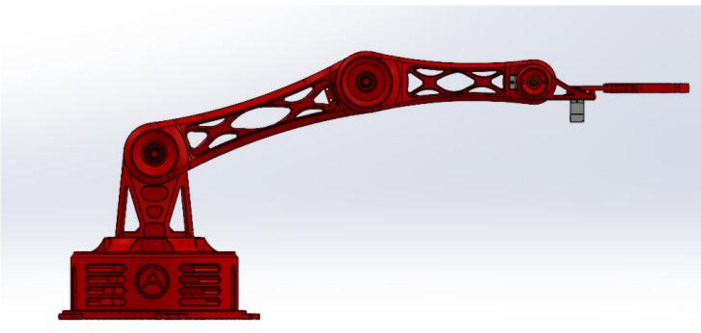

Figure 1: Fully Extended Arm

Figure 1 illustrates the loading assumptions used for the static analysis. Each component weight is represented as a concentrated force acting through its center of mass.

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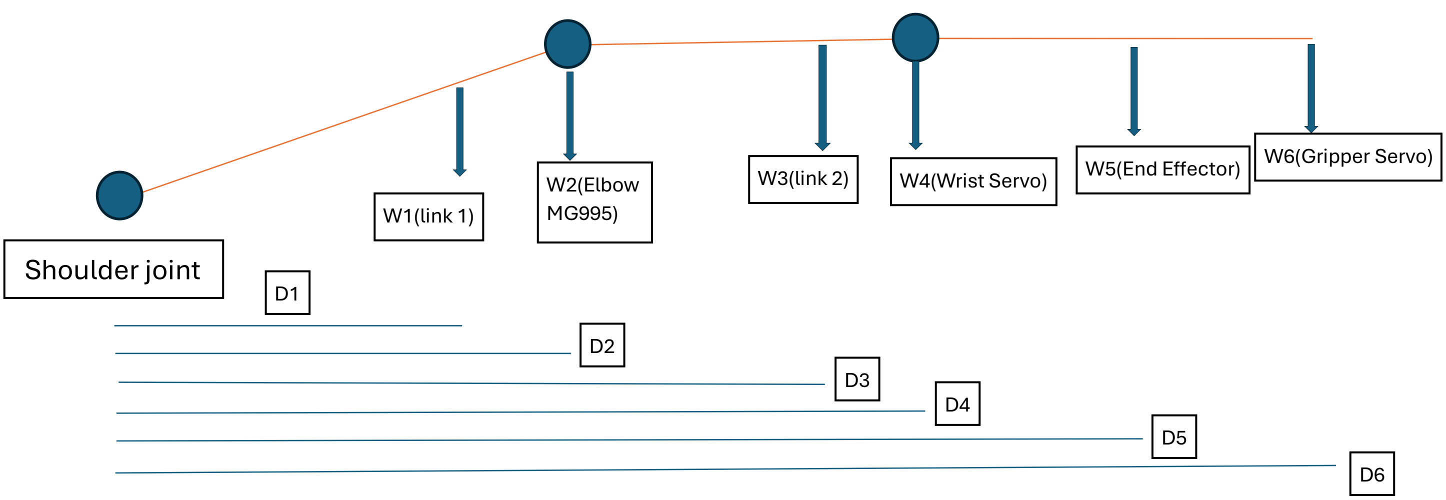

5. Determination of Moment Arms

The torque generated by each component depends on both its weight and the perpendicular distance from the shoulder joint to its center of mass. These distances were determined directly from the SolidWorks assembly using the maximum collision-limited extension of the Atlas V1 arm.

Each printed component was assigned the verified Anycubic PLA V3.0 material and its center of mass was obtained using the SolidWorks Mass Properties tool. Manufacturer specifications were used for the servo masses, while their center locations were approximated at the geometric centers of the servo bodies.

The maximum collision-limited extension was determined using the SolidWorks Collision Detection tool. Once this configuration was reached, the shoulder joint was fixed to preserve the arm position while the remaining measurements were obtained. This configuration is shown in Figure 1.

5.1 Link 1 Center of Mass

The Link 1 printed structure has a mass of **280.24 g**.

The center of mass was obtained directly from SolidWorks using the Mass Properties tool. A reference coordinate system was created at the shoulder axis to measure the horizontal distance from the shoulder joint to the Link 1 center of mass.

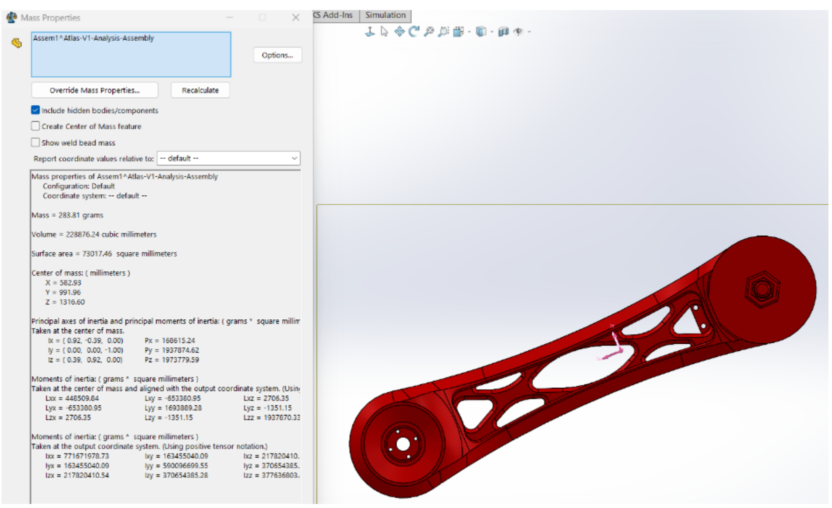

Figure 2: Link 1 Center of Mass and moment arm.

The maximum shoulder rotation was first determined using **SolidWorks Collision Detection**, allowing the arm to rotate until the first point of self-interference was reached. This configuration represents the maximum practical extension of Atlas V1 and was used as the worst-case loading condition for the static torque analysis.

After reaching this position, **Link 1 was fixed** to preserve the configuration. Construction centerlines were then created through the shoulder and elbow joint axes, and the angle of Link 1 relative to the horizontal reference was measured as **18.74°**.

The center of mass of Link 1 is located **120.03 mm** from the shoulder joint along the link centerline. Since gravity acts vertically downward, only the horizontal projection of this distance contributes to the moment arm about the shoulder joint. Therefore, the effective moment arm is calculated using

$$D_1 = r_1 \cos \theta_1$$

where

- $r_1 = 120.03$ mmis the distance from the shoulder axis to the Link 1 center of mass,
- $\theta_1 = 18.74°$is the measured angle between Link 1 and the horizontal.

Substituting the measured values,

$$D_1 = (120.03)\cos(18.74°)$$

$$\boxed{D_1 = 113.68 \text{ mm}}$$

This horizontal distance is subsequently used in the calculation of shoulder joint torque.

5.2 Elbow Servo

The elbow MG995 servo is mounted directly at the end of Link 1.

Using the assembly geometry,

$$D_2 = 189.50 \text{ mm}$$

5.3 Link 2 Center of Mass

The Link 2 printed structure has a mass of **164.24 g**.

The center of mass lies **72.44 mm** beyond the elbow axis.

Therefore,

$$D_3 = 189.50 + 72.44 = 261.94 \text{ mm}$$

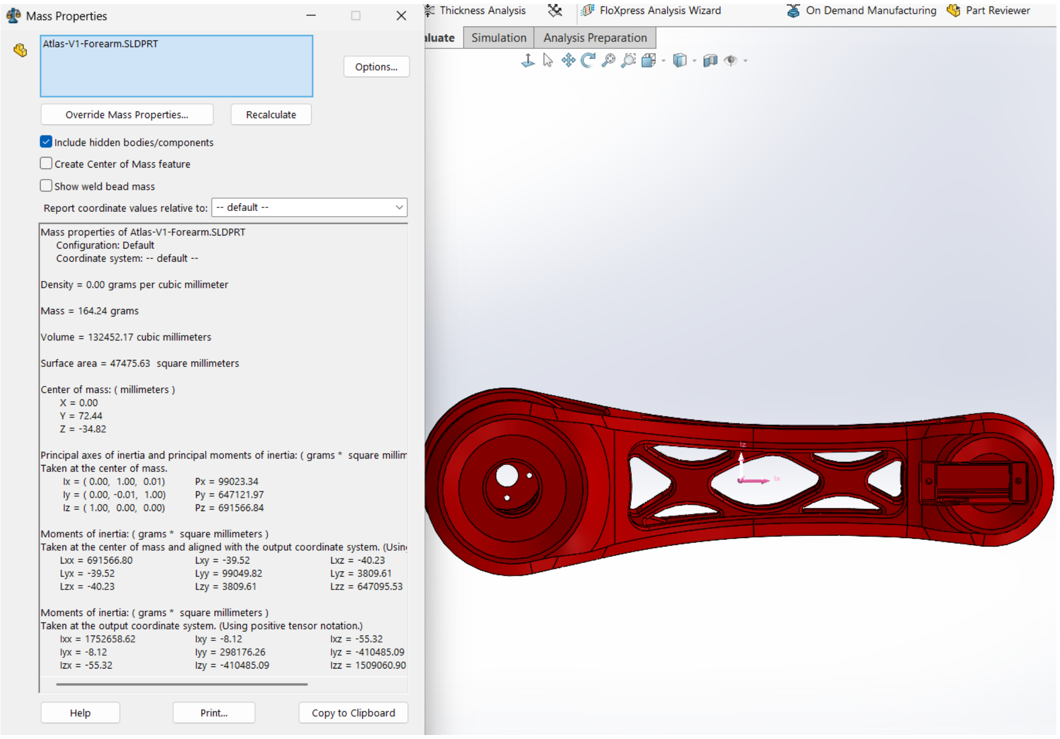

Figure 3: Link 2 Center of Mass and moment arm.

5.4 Wrist Servo

The wrist MG90S servo is located at the wrist joint.

Measured distance:

$$D_4 = 339.26 \text{ mm}$$

5.5 Printed End Effector

A local coordinate system was created with its origin at the wrist axis and the x-axis aligned with the gripper.

The SolidWorks Mass Properties tool reported

$$X = -50.73 \text{ mm}$$

Only the magnitude is required for the torque calculation.

Therefore,

$$D_5 = 339.26 + 50.73 = 389.99 \text{ mm}$$

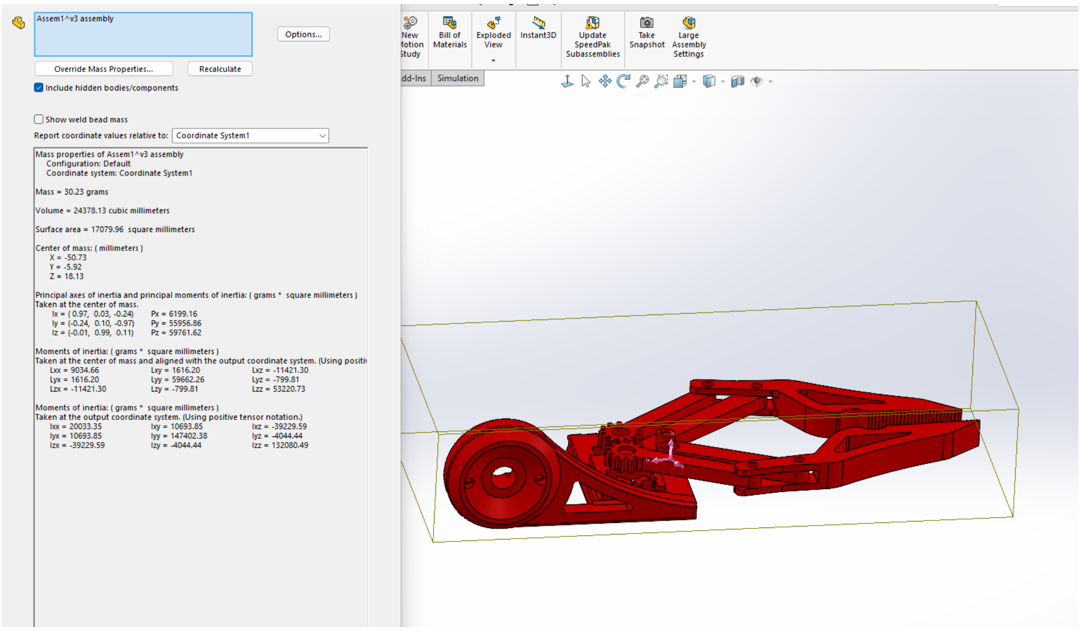

Figure 4: End Effector Center of Mass

5.6 Gripper Servo

The MG90S servo located within the gripper was measured directly in the assembly.

$$D_6 = 378.10 \text{ mm}$$

6. Shoulder Joint Static Torque Analysis

The static torque about the shoulder joint is obtained by summing the moment produced by each component.

For each component,

$$W_i = m_i g$$

and

$$\tau_i = W_i D_i$$

where

- $W_i$is the component weight,
- $D_i$is the horizontal distance from the shoulder axis to the component center of mass.

Link 1

Weight

$$W_1 = (0.28024)(9.81) = 2.749 \text{ N}$$

Moment arm

$$D_1 = 0.11368 \text{ m}$$

Torque

$$\tau_1 = (2.749)(0.11368) = 0.313 \text{ N} \cdot \text{m}$$

Elbow MG995

$$m_2 = 0.055 \text{ kg}$$
$$W_2 = (0.055)(9.81) = 0.540 \text{ N}$$
$$D_2 = 0.18950 \text{ m}$$
$$\tau_2 = (0.540)(0.18950) = 0.102 \text{ N} \cdot \text{m}$$

Link 2

$$m_3 = 0.16424 \text{ kg}$$
$$W_3 = (0.16424)(9.81) = 1.611 \text{ N}$$
$$D_3 = 0.26194 \text{ m}$$
$$\tau_3 = (1.611)(0.26194) = 0.422 \text{ N} \cdot \text{m}$$

Wrist Servo

$$m_4 = 0.0134 \text{ kg}$$
$$W_4 = (0.0134)(9.81) = 0.1315 \text{ N}$$
$$D_4 = 0.33926 \text{ m}$$
$$\tau_4 = (0.1315)(0.33926) = 0.0446 \text{ N} \cdot \text{m}$$

Printed End Effector

$$m_5 = 0.03023 \text{ kg}$$
$$W_5 = (0.03023)(9.81) = 0.2966 \text{ N}$$
$$D_5 = 0.38999 \text{ m}$$
$$\tau_5 = (0.2966)(0.38999) = 0.1157 \text{ N} \cdot \text{m}$$

Gripper Servo

$$m_6 = 0.0134 \text{ kg}$$
$$W_6 = (0.0134)(9.81) = 0.1315 \text{ N}$$

$$D_6 = 0.37810 \text{ m}$$
$$\tau_6 = (0.1315)(0.37810) = 0.0497 \text{ N} \cdot \text{m}$$

Total Shoulder Torque

$$T_{shoulder} = \sum_{i=1}^{6} \tau_i$$
$$T_{shoulder} = 0.313 + 0.102 + 0.422 + 0.0446 + 0.1157 + 0.0497$$
$$\boxed{T_{shoulder} = 1.047 \text{ N} \cdot \text{m}}$$

Converting to the units commonly used for servos,

$$T_{shoulder} = \frac{1.047}{0.09807} = 10.67 \text{ kg} \cdot \text{cm}$$

7. Results

The calculated static torque required at the shoulder joint is

$$T_{shoulder} = 1.047 \text{ N } \cdot \text{m}$$

which is equivalent to

$$10.67 \text{ kg} \cdot \text{m}$$

Servo Capability Comparison

The shoulder actuator used in Atlas V1 is the **Deegoo-FPV MG995 metal gear servo**. According to the manufacturer's published specification, the servo provides a **stall torque of 13 kg·cm at 6 V**.

The calculated static shoulder torque required by Atlas V1 is

$$T_{required} = 10.67 \ kg \cdot cm$$

Therefore,

$$10.67 < 13.00$$

The remaining static torque margin is

$$13.00 - 10.67 = 2.33\ kg \cdot cm$$

which corresponds to

$$\frac{2.33}{13.00} \times 100 = 17.9\%$$

The analysis therefore indicates that the selected MG995 servo is capable of supporting Atlas V1 under static loading without an external payload.

Discussion

The calculated shoulder torque represents the theoretical static holding torque required to support the Atlas V1 arm in its maximum collision-limited extension without any external payload.

This analysis assumes static equilibrium and neglects the effects of acceleration, friction, fasteners, wiring, and manufacturing tolerances. Consequently, the calculated torque should be considered the minimum required holding torque. In practical operation, additional torque margin is required to account for dynamic motion and ensure reliable performance.

Quantity	Value
Required shoulder torque	1.047 N·m
Required shoulder torque	10.67 kg·cm
MG995 specified stall torque (6 V)	13.00 kg·cm
Static utilization	82.1%
Remaining margin	17.9%

8. Conclusion

This analysis evaluated the static shoulder torque required to support the Atlas V1 robotic arm using SolidWorks mass properties and classical free-body analysis.

The maximum static shoulder torque was calculated to be

$$T_{shoulder} = 1.047\ N \cdot m$$

which is equivalent to

$$10.67\ kg \cdot cm.$$

Comparison with the manufacturer's specified stall torque of **13 kg·cm at 6 V** indicates that the selected Deegoo-FPV MG995 servo is capable of supporting the Atlas V1 arm in its maximum collision-limited configuration under static loading.

The calculated utilization is approximately **82%**, leaving a static torque margin of approximately **18%**. Although adequate for static operation, this margin does not account for dynamic effects such as acceleration, impact loading, payloads, friction, manufacturing tolerances, or long-term servo wear.

Future work will focus on:

- Static finite element analysis (FEA) of the printed components.
- Experimental validation using the assembled prototype.
- Dynamic torque analysis during motion.
- Evaluation of higher-performance actuators for future Atlas revisions.

References

1. Anycubic. *PLA V3.0 Technical Data Sheet.*
2. Dassault Systèmes. *SOLIDWORKS Mass Properties Documentation.*
3. Deegoo-FPV. MG995 Metal Gear Servo Motor Product Specifications (13 kg·cm stall torque at 6 V).